

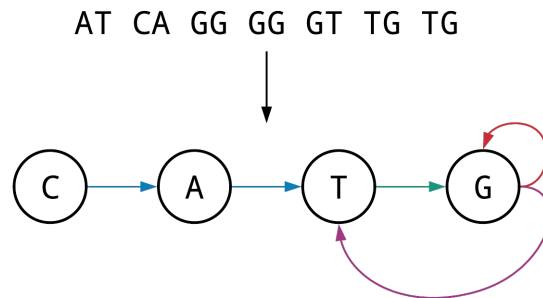
3K Generate Contigs from a Collection of Reads

Contig Generation Problem

Generate the contigs from a collection of reads (with imperfect coverage).

Input: A collection of k -mers *Patterns*.

Output: All contigs in the graph $\text{DEBRUIJN}(\text{Patterns})$.



Formatting

Input: A space-separated list of k -mer strings *Patterns*.

Output: A space-separated list of strings representing all contigs in the graph $\text{DEBRUIJN}(\text{Patterns})$ (you may return the strings in any order).

Constraints

- The number of strings in *Patterns* will be between 1 and 10^4 .
- The length of any given string in *Patterns* will be between 1 and 10^2 .

Test Cases

Case 1

Description: The sample dataset is not actually run on your code.

Input:

ATG ATG TGT TGG CAT GGA GAT AGA

Output:

CAT GAT TGT TGGG AGA ATG ATG

Case 2

Description: The sample dataset is not actually run on your code.

Input:

AG GT GC TA

Output:

GTAG GC

Case 3

Description: The sample dataset is not actually run on your code.

Input:

GTT TTA TAC TTT

Output:

GTT TTAC TTT

Case 4

Description: The sample dataset is not actually run on your code.

Input:

GAGA AGAG AACG ACGT ACGG

Output:

ACGT ACGG AACG GAGAG

Case 5

Description: The sample dataset is not actually run on your code.

Input:

TGAG GACT CTGA ACTG CTGA

Output:

CTGA CTGA TGAG GACTG

Case 6

Description: A larger dataset of the same size as that provided by the randomized autograder.